

Network Security Assessment - Nmap Scan Report

1. Introduction

This report presents the results of a network security assessment performed using Nmap (Network Mapper). The purpose of the scan was to identify network hosts, exposed services, open ports, and potential security weaknesses.

The assessment follows an IT audit approach by documenting technical observations, evaluating associated risks, and providing recommendations.

2. Assessment Overview

Item	Details
Assessment Type	Network Security Assessment
Tool Used	Nmap
Assessment Purpose	Identify exposed network services and security risks
Testing Approach	Non-intrusive security scanning
Assessment Status	Completed

3. Scan Objectives

The Nmap scan was performed to:

- Identify active network hosts.
 - Detect open ports.
 - Identify running services.
 - Review exposed network services.
 - Support vulnerability assessment activities.
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4. Scan Environment

Assessment Environment:

Controlled test environment for cybersecurity assessment purposes.

Target:

Authorized assessment system/network.

Note:

The assessment was performed in a controlled environment and did not include unauthorized scanning of external systems.

5. Nmap Scan Activities

5.1 Host Discovery

Purpose:

To identify active devices available within the assessment environment.

Expected Output:

- Active hosts
 - IP addresses
 - Network availability information
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5.2 Port Scanning

Purpose:

To identify accessible ports that may expose services.

Example findings:

Port	Service	Status
22	SSH	Open
80	HTTP	Open
443	HTTPS	Open
3389	Remote Desktop	Open

5.3 Service Detection

Purpose:

To identify services and versions running on open ports.

Example:

Service	Risk Consideration
SSH	Requires secure configuration
HTTP	Requires secure web configuration
Remote Desktop	Requires strong access controls

6. Scan Findings Summary

Finding ID	Observation	Risk Level
NMAP-001	Multiple open ports identified	Medium
NMAP-002	Remote access service exposed	High
NMAP-003	Services require security configuration review	Medium

7. Security Analysis

Finding 1: Exposed Network Services

Risk Level: Medium

Observation:

Multiple network services were identified as accessible.

Potential Impact:

Unnecessary exposed services may increase the attack surface.

Recommendation:

- Disable unnecessary services.
 - Review firewall rules.
 - Restrict access based on business requirements.
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Finding 2: Remote Access Service Exposure

Risk Level: High

Observation:

Remote access services require careful security management.

Potential Impact:

Weak authentication or misconfiguration may allow unauthorized access attempts.

Recommendation:

- Implement multi-factor authentication.
 - Restrict remote access.
 - Monitor login activities.
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8. Risk Assessment Summary

Category	Rating
Network Exposure	Medium
Service Configuration	Medium
Remote Access Security	High

9. Recommendations

Recommended security improvements:

- Perform regular network vulnerability assessments.
 - Remove unnecessary open ports.
 - Maintain updated service versions.
 - Apply firewall restrictions.
 - Monitor network activity.
 - Review access control policies.
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10. Conclusion

The Nmap-based Network Security Assessment identified potential areas where network security controls can be strengthened. The findings demonstrate the importance of regular network monitoring, vulnerability assessment, and proactive risk management.